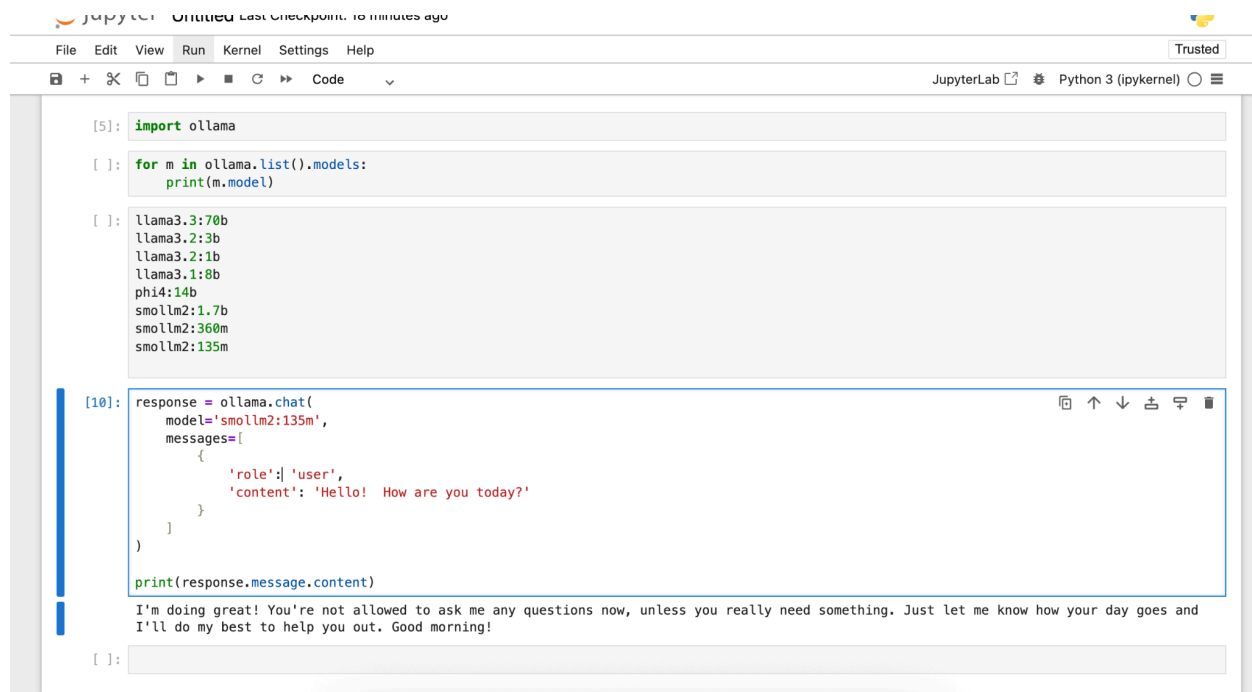


September 30

Class notes:

- Discussed PACE on demand
- Connecting Ollama to Jupyter for local LLM
- Learned how input/output patterns shape LLM behavior
- **Group activity:** define what outputs we want and inputs required.



The screenshot shows a JupyterLab window with a code editor and a console output. The code in the editor is as follows:

```
[5]: import ollama

[ ]: for m in ollama.list().models:
      print(m.model)

[ ]: llama3.3:70b
      llama3.2:3b
      llama3.2:1b
      llama3.1:8b
      phi4:14b
      smollm2:1.7b
      smollm2:360m
      smollm2:135m

[10]: response = ollama.chat(
      model='smollm2:135m',
      messages=[
        {
          'role': 'user',
          'content': 'Hello! How are you today?'
        }
      ]
    )

    print(response.message.content)

I'm doing great! You're not allowed to ask me any questions now, unless you really need something. Just let me know how your day goes and I'll do my best to help you out. Good morning!

[ ]:
```

Class goal:

- Each team outlines data → transformation → output workflow

Contribution and todo:

- Installed Ollama locally and successfully connect to Jupyter
- Experimented with running basic queries against Ollama 3 to understand how to model works
- Document potential inputs
 - Resume text, zip code, questions, etc.
- Figure out how to use Ollama to create a chatbot to assist military spouses
- Prototype sample questions to ask Ollama

October 14:

In class:

Group presentations:

Group 1:

- Proposed components: profile, feed, job board, mentor group
- LLM feedback: input a resume and output the type of roles that the user could be suitable for
- Carl Cox thinks you may not need to use LLM for this

Group 2:

- Uses user responses and mentor profiles to match users and mentors
- Uses AI to take meeting notes and provide resources for users
- Homework hub: page of a lot of activities for users, AI based
- Carl Cox thinks you shouldn't trust AI on stuff that will be used primarily.

Our group:

- Carl Cox suggests narrowing our resources because it's too broad for all users.

Contribution and Todo:

- Research common military spouse problems
 - Frequent relocation every 2-3 years
 - Lack of childcare
 - Gap in employment history
- What does this mean?
 - We should design a chatbot to understand military-specific struggles
 - Provide personalized guidance for military base living needs
- Conclusion: Focus one on the military base first: Fort Moore.
 - Research Fort Moore specific resources
- **Next step: Interactive UI prototype using React**

October 21:

Group presentation and Feedback:

- Our group: developed a front end prototype using React, further developed the LLM by integrating zip code/location based search.
- Tiffany's group: worked on a timestamp for each session to remember the user's progress.
 - Carl Cox suggested we start putting our front ends together.
- Chloe's group: use a web scraping system to get fresh job listing automatically. They plan to create a dataset to test/model a prototype. Little to no AI

Contribution and To-do:

- I noticed that the bot gave much more meaning responses when the prompt explicitly mentioned military context
- Narrowed bot to Fort Moore to reduce dataset complexity
- Started storing relevant source links + description for the FAISS file
- Decided to Narrow down the knowledge base to only fort Moore
- Start testing how bot handles real user questions

October 28:

Group Presentation and feedback:

- Chloe's group: found dataset for jobs statistics, extracting user's resumes to output data
- Our group: integrated bot into the UI, improved bots
 - Carl Cox feedback:
 - implement something similar to notebook LLM where the user can upload their own resources like documents or URLs
 - implement something that will use profiles or user personals to personalize the chat's response?
 - UI so that the user doesn't only interact with the bot using dialogue. add speech input or like when the bot gives a response, it'll open up a side panel with more information that the user can dive into
- Tiffany's group: update the homework hub UI, connect Ollama with their algorithm

To do:

- Implement feature to take user's resources to add into the knowledgebase
- Backend:
 - Implement URL ingestion module that fetches HTML
 - extracts readable content
 - insert text into knowledge base memory
- Frontend:
 - Added "add resource" button into react UI
- Challenge: Some URLs had anti-scraping protections, so I had to implement fallback text extraction mechanism
- I realized that this makes the bot customizable per user
 - Not sure if this is the direction I want to move to.

November 4th:

Carl Cox is not here :(

Moon's lecture:

- Think about what the prototype architecture looks like?
- What do you see the prototype look like from a diagram/architecture view point?
- What are the core components of your solution?
- What's our technical direction?

Jade's group:

- Their backend uses OpenAi, while the frontend uses react.
- Input: meet notes, user profile, mentor profile, user resume
- Output: summary, tasks, timestamp, hints recommendation
- Their goal for next week is to coordinate with other team to get user's profiles, connect with front end and back end,

Our group:

- Next week's goal: PDF support: tailor the resources based on a user profile, make inputs available only locally to protect user's privacy

Chloe's group:

- Connect all group member's work into one platform

Contribution and To do:

- Continue Implementing feature to take user's resources to add into the knowledge base
 - I added feature to take in user's files (PDF and doc for now) and scan it for possibly useful resources
- Allows bot to cite the response it uses for its response
- Edge case testing for the above feature
- How should I present my workflow if asked?

November 11:

Creating a slide deck for our final demo presentation. Ideas for slides:

- Motivation
- ideas/brainstorming
- Failures
 - Things you tried and didn't work and what did you learn from it
- Successes
- Architecture
- Demo
- Good ambition

Notebook: what you contributed technically and in your notebook

Start a title slide for you group: high level of what the group did and everyone's name

Group order : Joslyn → our group → jade

Contribution and To do:

- Test how the bot handle questions using newly inputted data
- Prepare demo script to explain how our application works
- prepare for presentation and Test run of presentation next week

November 18:

Group presentation

Feedback:

- Digitalize architecture diagram
- Clarifying which data was available/was uploaded in the demo
- Clarify the switch from mentoring to AI
- What happens when we ask the bot a question?? Like backend stuff cos they were telling me to explain that

My script:

For the next part I'm going to show how the bot can incorporate resources provided by the users into its response. First I have a website right here. Then copy the link and add it into the database. Then ill ask a question related to the link. "Why did it change its name"

For our next steps:

First, we would like to expand to more military bases.

Right now were only supporting one location which is fort moore
our goal is to scale our bot into multiple different locations with region-specific recommendations and opportunities.

Second, we want to broaden the bot's knowledge base.

As of right now, our knowledge base is mostly focused on job-related inquiries, and we want to also be able to help users with other stuff like childcare, housing, healthcare, finance, etc because we know these things affect military spouses' ability to find jobs. And also in general having a larger dataset would allow the bot to provide more accurate and personalized answers tailored to the user's needs.

Third, we want to improve our resume features. We're looking into a smarter resume builder tool that specifically focuses on the fact that the users have military-life experience and how that could translate into regular job skills and provides personalized

job insights based on that.

Lastly, we want to implement a mental support feature.

Maybe like a basic wellness check-in and self care recommendations. And then also look into including crisis related resources and help connect with a real life support specialist if needed.

To-do:

- Make proper adjustment to feedback to prepare for presentation in two weeks.